Divyam Samarwal

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Education

Netaji Subhas University of Technology, New Delhi Aug 2024 – May 2028
Bachelor of Technology, Mathematics and Computing **GPA:** 7.09/10

- Relevant Coursework: Probability & Statistics, Applied Linear Algebra, Design & Analysis of Algorithms, Optimization Techniques, Machine Learning, Database Management Systems, Theory of Automata & Formal Languages

Experience

WorldQuant BRAIN Remote
Quantitative Research Consultant Aug 2025 – Present

- Achieved the Gold Level in the Global Alpha Challenge with a score of 12,225+, ranking in the top global percentile in 50+ countries.
- Developed and backtested 20+ alpha factors in price-volume data; applied cross-sectional neutralization and factor decay analysis to achieve an annualized Sharpe ratio of 1.4+ with controlled systematic risk.

Projects

Paper Trading Dashboard *Next.js, TypeScript, WebSockets, SQLite, PostgreSQL* Jan 2026
github.com/DivyamSamarwal/paper-trading

- Engineered a real-time paper trading engine with WebSocket-driven feeds, processing 1,000+ simulated trades per minute with sub-50ms execution latency while supporting 100+ concurrent users on a global leaderboard.
- Developed a price simulation engine using Gaussian random walk at 1,000ms ticks, backed by a hybrid SQLite/PostgreSQL database that enabled zero-change local-to-production deployment and handled 10,000+ simulated data points daily.

CFG Converter & Visualizer *TypeScript, Next.js, Algorithms, Tesseract.js* Feb 2026
divyamsamarwal.github.io/cfg/

- Engineered an automata toolkit to transform Context-Free Grammars into Chomsky and Greibach normal forms, reducing manual derivation time by 90% and successfully validating 500+ complex production rules.
- Implemented the CYK Algorithm and integrated Tesseract.js OCR, processing multimodal inputs (image/voice) in under 2 seconds and rendering real-time visual parsing tables for strings up to length 50.

PyOpenF1 *Python, REST APIs, Pandas, Data Engineering* Apr 2026
github.com/DivyamSamarwal/pyopenf1

- Developed an open-source Python wrapper for the OpenF1 API, efficiently ingesting and structuring high-frequency Formula 1 telemetry and timing data streaming at 4Hz across 20+ cars per session.
- Engineered robust data pipelines to transform 100,000+ rows of raw JSON streams per race into structured Pandas DataFrames, decreasing data retrieval and preprocessing time by 60% for large-scale time-series analysis.

Autobot *Python, discord.py, asyncio, Flask* Jun 2021
techsaurus.vercel.app

- Developed an asynchronous Discord bot using Python and Flask, successfully scaling to serve 50+ communities and processing over 10,000 daily user commands with 99.9% uptime.
- Integrated 5+ third-party APIs (weather, social media, anime databases) to deliver dynamic content, reducing server response times by 30% through optimized asyncio event loops.

Technical Skills

Programming Languages: Python · TypeScript · JavaScript · C++ · Rust · SQL · HTML/CSS
Libraries and Tools: Pandas · NumPy · Next.js · WebSockets · PostgreSQL · SQLite · React · Flask · Git · MATLAB · LaTeX

Honors and Awards

WorldQuant Global Alpha Challenge – Gold Level, 12,225+ points, top global percentile 2025
Trinity College London, Plectrum Guitar Grade 1 – 97/100, Distinction 2019